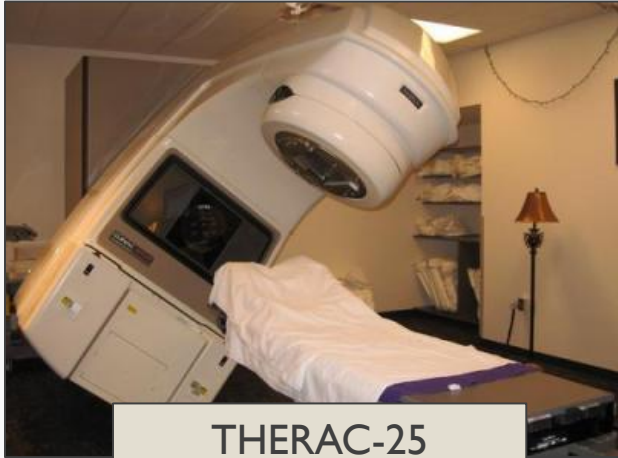


A Monitoring-Oriented Theory and Classification of Quantitative Specifications

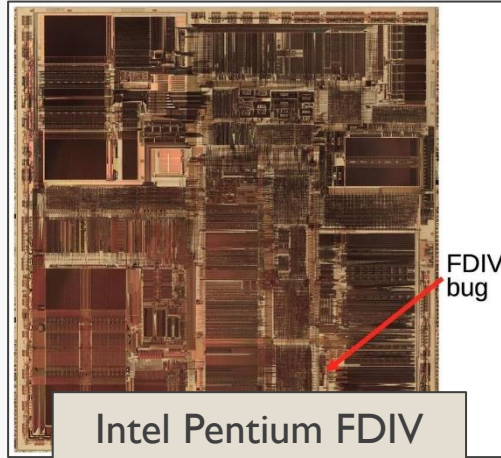
N. Ege Saraç



Trustworthy Software Systems?



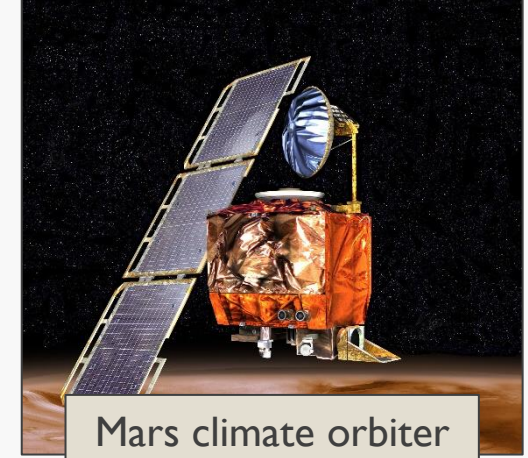
THERAC-25
(1985)



Intel Pentium FDIV
(1994)



Ariane flight V88
(1996)



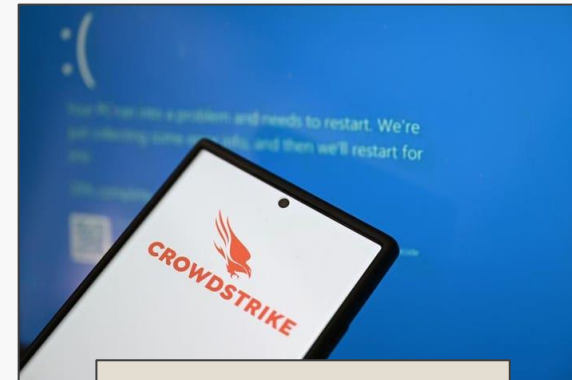
Mars climate orbiter
(1999)



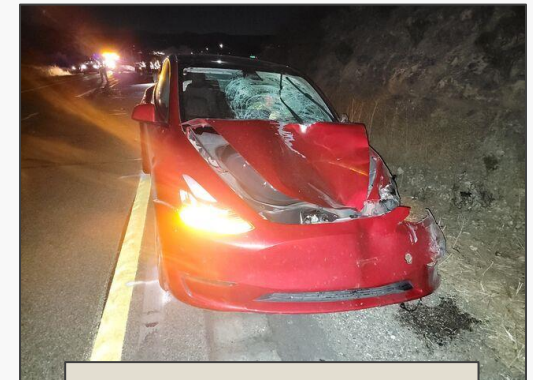
Northeast blackout
(2003)



Knight trading glitch
(2012)

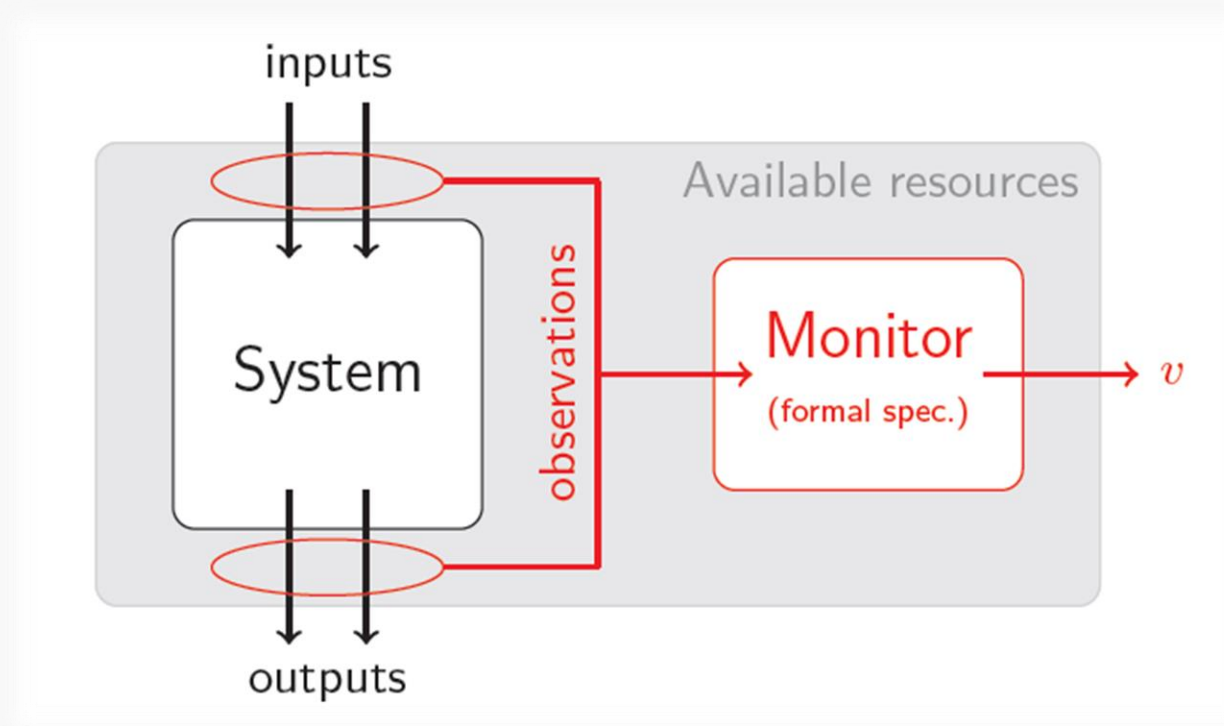


CrowdStrike outage
(2024)



Self-driving crashes
(2025)

Online Black-Box Monitoring



- Monitor runs in **parallel** and outputs a **stream of verdicts**
- Monitor has **no knowledge** of the system's internals
- Monitor **shares resources** with the system

Why Monitoring Should Become Ubiquitous

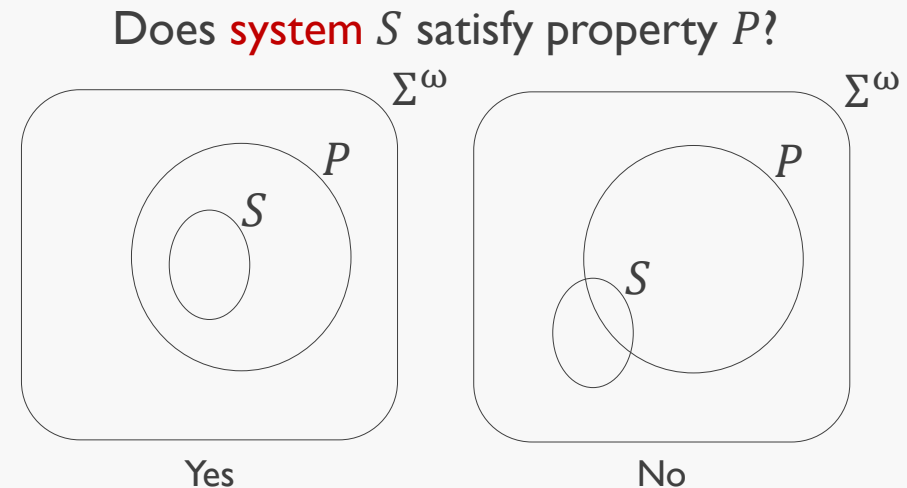
- Increasing **software complexity**
 - Verification gap
- Increasing **hardware parallelism**
 - Some resources into assurance
- **Inclusion** vs. **membership**
 - More expressive formalisms

Why Monitoring Should Become Ubiquitous

- Increasing **software complexity**
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Why Monitoring Should Become Ubiquitous

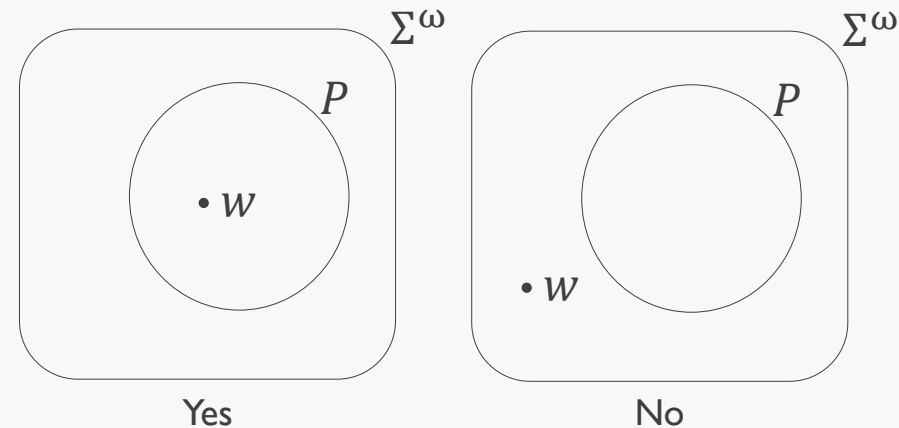
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Why Monitoring Should Become Ubiquitous

- Increasing **software complexity**
 - Verification gap
- Increasing **hardware parallelism**
 - Some resources into assurance
- **Inclusion** vs. **membership**
 - More expressive formalisms

Does **system behavior** w satisfy property P ?



Why Monitoring Should be Quantitative

- More **interesting** specifications
 - Response time, power consumption...
- Fully **universal** monitoring
 - Best possible estimates at all times
- More fitting for **approximate** methods
 - Trading monitor precision against monitor resources

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- More **interesting** specifications
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Contributions and Overview

1. A theory of quantitative **monitorability** for reasoning about **resource-precision tradeoffs**
2. A **monitoring-oriented classification** of quantitative properties
3. A **software tool** for analyzing quantitative automata

Specifying System Properties

System behaviors (a.k.a. words, traces, executions).

- $\Sigma = \{r, g, t, o\}$ Σ^ω : infinite words over Σ Σ^* : finite words over Σ
- $w = t t (r t g)^\omega$

Boolean property.

- $P \subseteq \Sigma^\omega$
- “Is every r eventually followed by g ?”

Quantitative property.

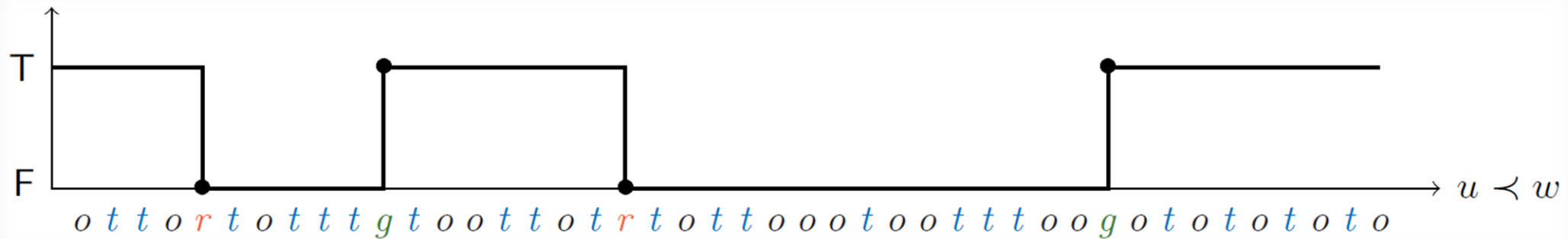
- $\Phi : \Sigma^\omega \rightarrow \mathbb{D}$ where $\mathbb{D}$ is a value domain (e.g., real numbers)
- “What is the average number of t ’s between each r and the matching g ?”

Part I

Quantitative and Approximate Monitoring

What and Why of Limit Monitoring

- $\Sigma = \{r, g, t, o\}$
- $P =$ every r is followed by some g



Limit monitoring

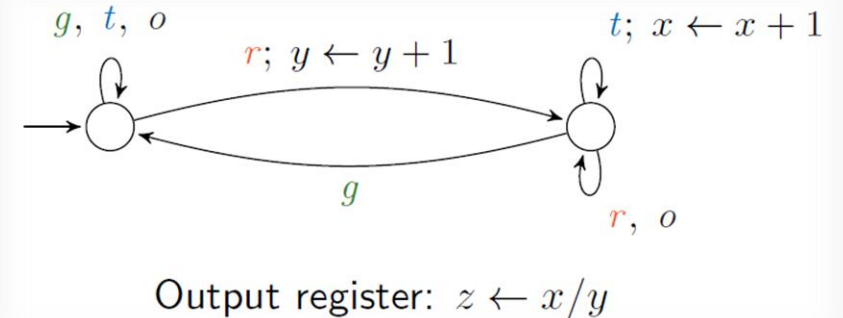
- Cares about the **tail** of monitor's estimates
- Can be **speculative**, **universal**, and **quantitative**

Quantitative Limit Monitoring with Register Monitors

Example (average response-time).

- $\Sigma = \{r, g, t, o\}$
- $\Phi(w) =$ lower limit of the running avg. of the occurrences of t between each r and the matching g in w

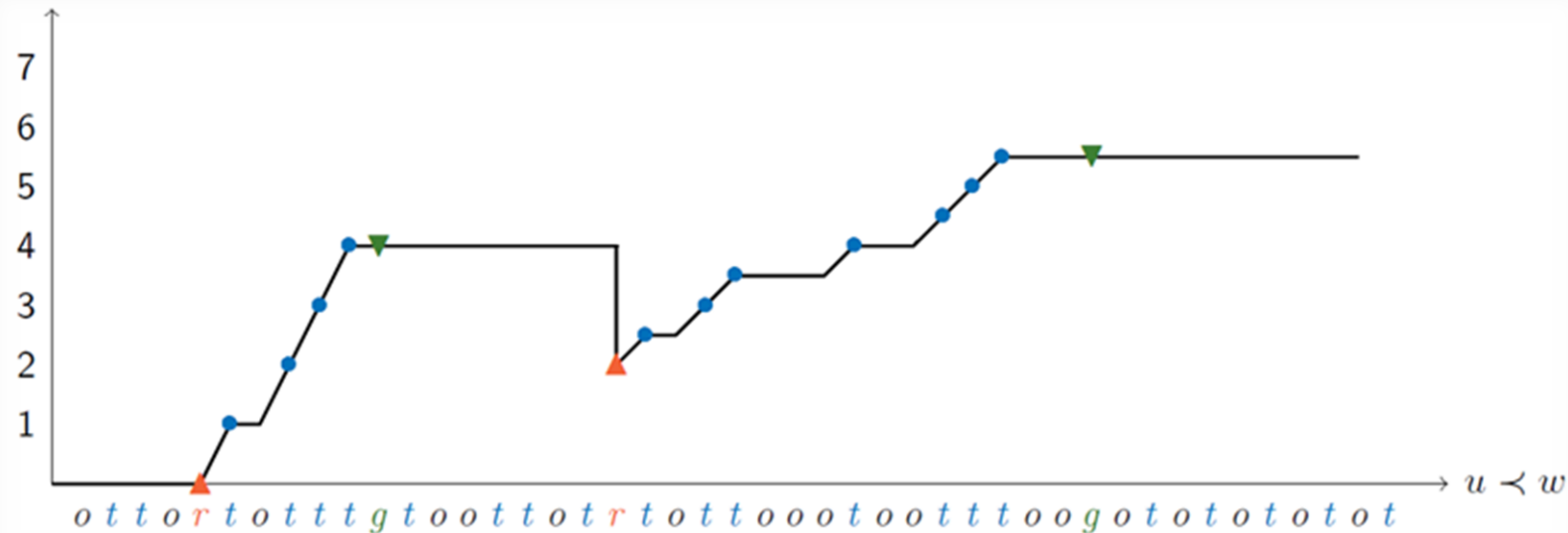
- 3 registers with “increment” and “divide”
- Non-monotonic



Quantitative Limit Monitoring with Register Monitors

Example (average response-time).

- $\Sigma = \{r, g, t, o\}$
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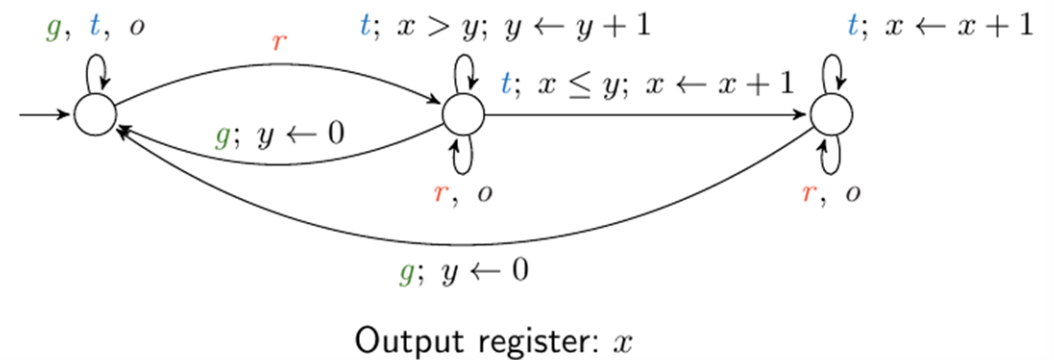


Quantitative Limit Monitoring with Register Monitors

Example (maximal response-time).

- $\Sigma = \{r, g, t, o\}$
- $\Phi(w) =$ least upper bound on the occurrences of t between each r and the matching g in w

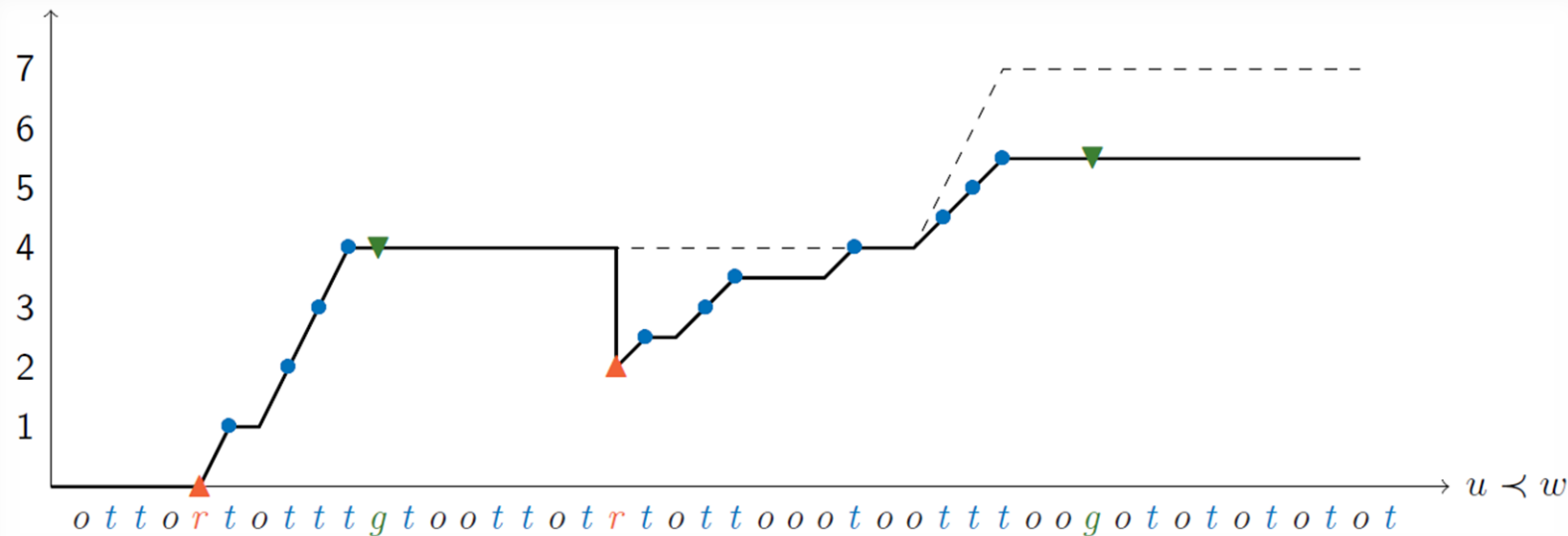
- 2 registers with “increment” (i.e., counters)
- Monotonic



Quantitative Limit Monitoring with Register Monitors

Example (maximal response-time).

- $\Sigma = \{r, g, t, o\}$
- $\Phi(w) =$ least upper bound on the occurrences of t between each r and the matching g in w



Quantitative Limit Monitorability

For property $\Phi: \Sigma^\omega \rightarrow \mathbb{D}$ and monitors $M, M': \Sigma^* \rightarrow \mathbb{D}$

Φ is **universally** monitorable from below by M iff

- $\forall w \in \Sigma^\omega: \limsup_{u < w} M(u) = \Phi(w)$

Φ is **approximately** monitorable from below by M iff

- $\forall w \in \Sigma^\omega: \limsup_{u < w} M(u) \leq \Phi(w)$

M' is **more precise** than M iff

- $\forall w: \limsup_{u < w} M(u) \leq \limsup_{u < w} M'(u)$
- $\exists w': \limsup_{u < w'} M(u) < \limsup_{u < w'} M'(u)$

An Infinite Hierarchy of Register Monitors

Theorem.

There is a quantitative property Φ such that

- Φ is not universally monitorable by any counter monitor
- each additional counter provides more precision

What and Why of Prompt Monitoring

How “informative” is a limit monitor on finite traces?

Prompt monitoring

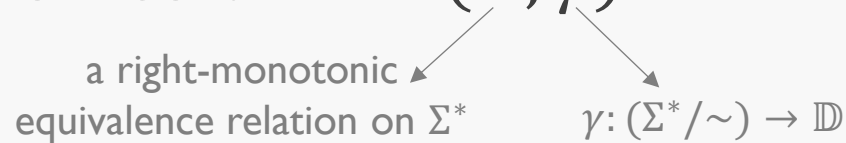
- Cares about monitor’s **finitary** estimates
- Can also be **speculative**, **universal**, and **quantitative**

Quantitative Prompt Monitoring with Abstract Monitors

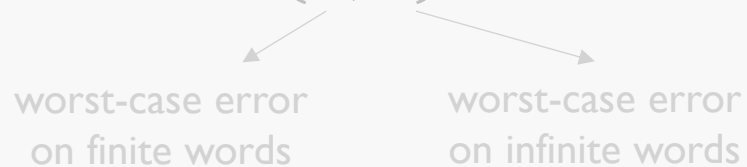
Limit property. $\Phi = (\pi, \text{Val})$



Abstract monitor. $M = (\sim, \gamma)$



- Resource use of M = how fast $\sim$ grows
- “ M is a (δ, ε) -monitor for Φ ”



$$\Phi(u) = \pi(u)$$

$$\Phi(w) = \text{Val}_{u < w} \pi(u)$$

$$M(u) = \gamma([u])$$

$$M(w) = \text{Val}_{u < w} \gamma([u])$$

$$\delta = \sup_{u \in \Sigma^*} |\Phi(u) - M(u)|$$

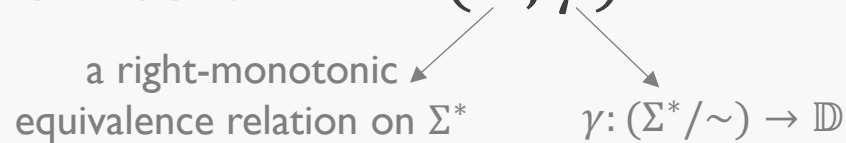
$$\varepsilon = \sup_{w \in \Sigma^\omega} |\Phi(w) - M(w)|$$

Quantitative Prompt Monitoring with Abstract Monitors

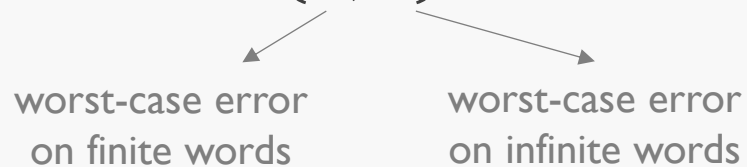
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$$\Phi(u) = \pi(u)$$

$$\Phi(w) = \text{Val}_{u < w} \pi(u)$$

$$M(u) = \gamma([u])$$

$$M(w) = \text{Val}_{u < w} \gamma([u])$$

$$\delta = \sup_{u \in \Sigma^*} |\Phi(u) - M(u)|$$

$$\varepsilon = \sup_{w \in \Sigma^\omega} |\Phi(w) - M(w)|$$

An Infinite Hierarchy of Abstract Monitors

Theorem.

Consider the maximal response-time property $\Phi_{\max}$.

For all $\delta \in \mathbb{N}$ there is a monitor M_δ such that

- M_δ is a (δ, δ) -monitor for $\Phi_{\max}$
- M_δ uses fewer resources than M_ε for all $\varepsilon < \delta$

Part II

Safety and Liveness of Quantitative Properties and Automata

Henzinger, Mazzocchi, and Saraç, “Quantitative Safety and Liveness”, FoSSaCS 2023.

Boker, Henzinger, Mazzocchi, and Saraç, “Safety and Liveness of Quantitative Automata”, CONCUR 2023.

Boker, Henzinger, Mazzocchi, and Saraç, “Safety and Liveness of Quantitative Properties and Automata”, LMCS Vol. 21 Iss. 2, 2025.

Boolean Safety

Example (no double request).

- $\Sigma = \{r, g, t, o\}$
- $P =$ no r is followed by another r without some g in between
 $w = t r t o t t o g t o o r t t o r t t o g t t \dots$

A boolean property $P : \Sigma^\omega \rightarrow \{0,1\}$ is *safe* iff
 $\forall w \in \Sigma^\omega : P(w) \neq 1 \Rightarrow \exists u < w : \forall w' \in \Sigma^\omega : P(uw') \neq 1$
For every word, wrong membership hypotheses have a finite witness.

Boolean Safety

Example (no double request).

- $\Sigma = \{r, g, t, o\}$
- $P =$ no r is followed by another r without some g in between

$w = t r t o t t o g t o o r t t o r t t o g t t \dots$

$M_P : 1 \dots \dots \dots 0 \dots \dots \dots$

A boolean property $P : \Sigma^\omega \rightarrow \{0,1\}$ is *safe* iff
 $\forall w \in \Sigma^\omega : P(w) \neq 1 \Rightarrow \exists u < w : \forall w' \in \Sigma^\omega : P(uw') \neq 1$

For every word, wrong membership hypotheses have a finite witness.

Quantitative Safety

Example (minimal response time).

- $\Sigma = \{r, g, t, o\}$
- $\Phi(w)$ = greatest lower bound on the occurrences of t between each r and the matching g in w

$w = t r t o t t o g t o o r t t o r t t o g t t \dots$

$M_{\Phi \geq 3} : T \dots \dots \dots F \dots \dots$

A quantitative property $\Phi : \Sigma^\omega \rightarrow \mathbb{D}$ is *safe* iff
 $\forall w \in \Sigma^\omega : \forall v \in \mathbb{D} : \Phi(w) \not\geq v \Rightarrow \exists u < w : \sup_{w' \in \Sigma^\omega} \Phi(uw') \not\geq v$

For every word, **every** wrong membership hypothesis has a finite witness.

Quantitative Safety Closure

Example (minimal response time).

- $\Sigma = \{r, g, t, o\}$
- $\Phi(w)$ = greatest lower bound on the occurrences of t between each r and the matching g in w
 $w = t r t o t t o g t o o r t t o r t t o g t t \dots$
sup : $\infty \dots \dots \dots 3 \dots \dots \dots 2 \dots \dots$

$$\Phi^*(w) = \inf_{u < w} \sup_{w' \in \Sigma^\omega} \Phi(uw')$$

Make Φ safe by increasing each word's value as little as possible.

Characterizing Quantitative Safety

Theorem.

- Φ^* is the least safety property that bounds Φ from above
- Φ is safe iff $\Phi = \Phi^*$
- Φ is safe iff Φ is upper semicontinuous (when $\mathbb{D}$ is a total order)
- Φ is safe iff Φ is universally monitorable by a nonincreasing monitor

$$\Phi^*(w) = \inf_{u < w} \sup_{w' \in \Sigma^\omega} \Phi(uw')$$

Make Φ safe by increasing each word's value as little as possible.

Quantitative Co-safety

Example (maximal response time).

- $\Sigma = \{r, g, t, o\}$
- $\Phi(w)$ = least upper bound on the occurrences of t between each r and the matching g in w

$w = t r t o t t o g t o o r t t o r t t o g t t \dots$

$M_{\Phi \leq 3} : T \dots \dots \dots F \dots \dots \dots$

$\text{inf} : 0 \dots 1 \dots 2.3 \dots \dots \dots 4 \dots \dots \dots$

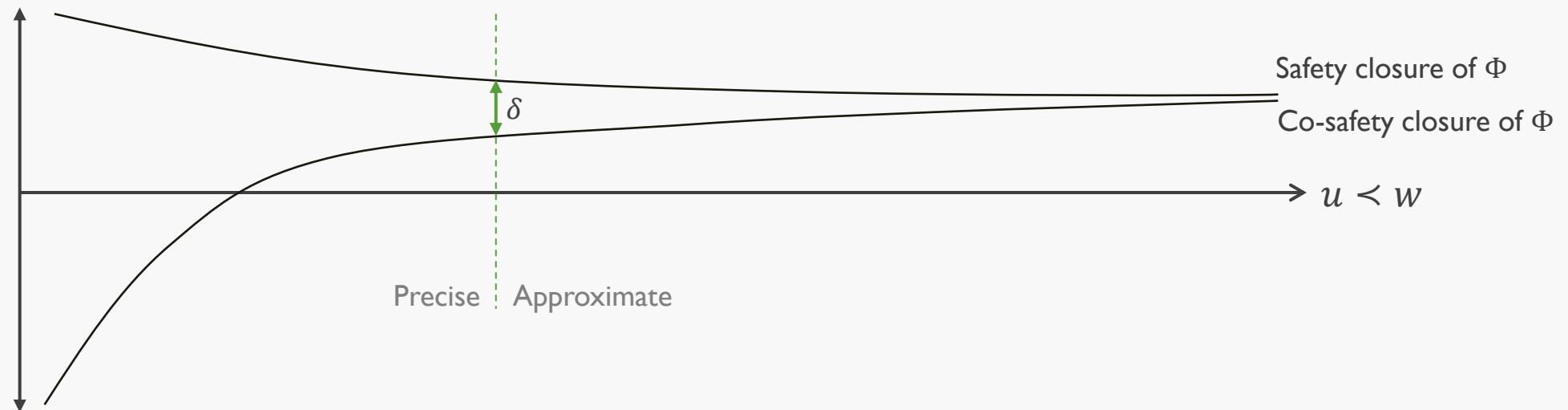
A quantitative property $\Phi : \Sigma^\omega \rightarrow \mathbb{D}$ is *co-safe* iff
 $\forall w \in \Sigma^\omega : \forall v \in \mathbb{D} : \Phi(w) \not\leq v \Rightarrow \exists u < w : \inf_{w' \in \Sigma^\omega} \Phi(uw') \not\leq v$

Finite-State Approximate Monitoring

Theorem.

Φ is safe and co-safe
implies

for every $\delta > 0$ there is a *finite-state* (δ, δ) -monitor for Φ



Boolean Liveness

Example (response).

- $\Sigma = \{r, g, t, o\}$

- P = every r is eventually followed by some g

$w = t r t o t t o g t o o r t t o r t t o g t t \dots$

$M_P : T \dots\dots\dots$

A boolean property $P: \Sigma^\omega \rightarrow \{0,1\}$ is *live* iff

$$\forall w \in \Sigma^\omega : P(w) \not\geq 1 \Rightarrow \forall u < w : \exists w' \in \Sigma^\omega : P(uw') \geq 1$$

For every word, wrong membership hypotheses have no finite witness.

Quantitative Liveness

Example (average response time).

- $\Sigma = \{r, g, t, o\}$
- $\Phi(w)$ = lower limit of the running avg. of the occurrences of t between each r and the matching g in w

$w = t r t o t t o g t o o r t t o r t t o g t t \dots$

$M_{\Phi \geq 3} : T \dots\dots\dots$

A quantitative property $\Phi : \Sigma^\omega \rightarrow \mathbb{D}$ is *live* iff

$$\forall w \in \Sigma^\omega : \Phi(w) < T \Rightarrow \exists v \in \mathbb{D} : \Phi(w) \not\geq v \wedge \forall u < w : \sup_{w' \in \Sigma^\omega} \Phi(uw') \geq v$$

For every word (with value $< T$), **some** wrong membership hypothesis has no finite witness.

Quantitative Liveness

Theorem.

Φ is live

iff

$\Phi(w) < \Phi^*(w)$ whenever $\Phi(w) < T$

Example (average response time).

- $\Sigma = \{r, g, t, o\}$
- $\Phi(w)$ = lower limit of the running avg. of the occurrences of t between each r and the matching g in w

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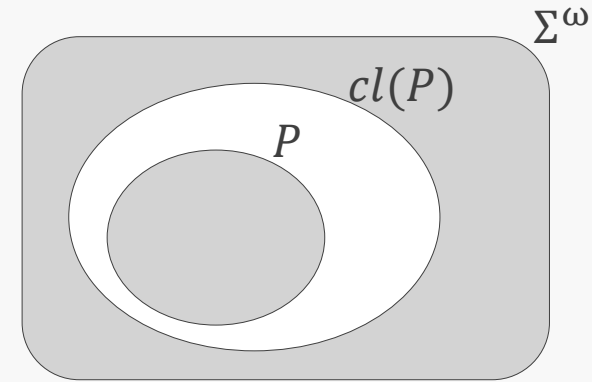
For every word (with value $< T$), **some** wrong membership hypothesis has no finite witness.

Safety-Liveness Decomposition

In the boolean setting.

$$\bullet P = \underbrace{cl(P)}_{\text{safe}} \cap \underbrace{\left(P \cup \left(\Sigma^\omega \setminus cl(P) \right) \right)}_{\text{live}}$$

Alpern and Schneider, “Defining liveness”, 1985.



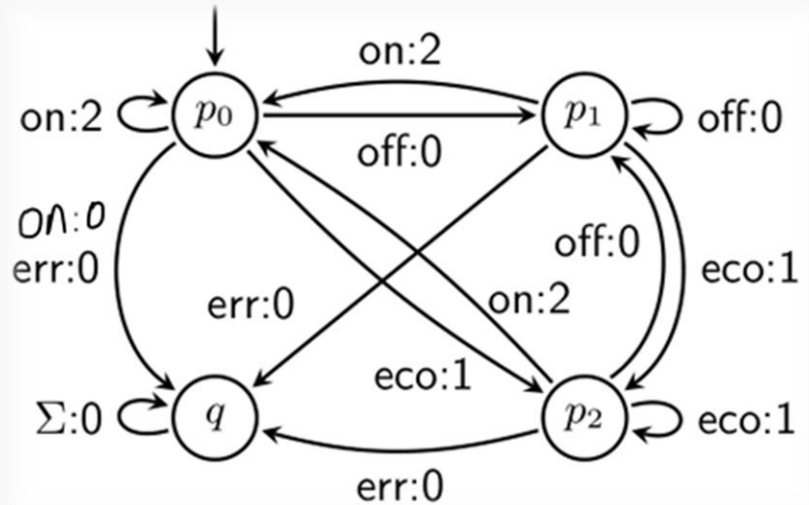
In the quantitative setting.

Theorem.

$$\Phi = \min(\underbrace{\Phi^*}_{\text{safe}}, \underbrace{\Psi}_{\text{live}})$$

$$\Psi(w) = \begin{cases} \Phi(w) & \text{if } \Phi(w) < \Phi^*(w) \\ \top & \text{if } \Phi(w) = \Phi^*(w) \end{cases}$$

Quantitative Automata



Quantitative automata are well-behaved:

- Finite-state
- Totally-ordered $\mathbb{D}$
- Bounded
- Supremum = maximum

- Acceptance condition $\rightarrow$ Value function
- $\text{Val} \in \{\text{Inf}, \text{Sup}, \text{LimInf}, \text{LimSup}, \text{LimInfAvg}, \text{LimSupAvg}, \text{DSum}\}$

Safety

Reachability

Co-Büchi

Büchi

Games with quantitative objectives

What's Decidable About These Automata?

	Inf	Sup	LimInf	LimSup	LimInfAvg	LimSupAvg	DSum
Emptiness (w.r.t v) $\exists w: A(w) \geq v$	PTime						
Universality (w.r.t v) $\forall w: A(w) \geq v$	PSpace-complete				Undecidable		Open
Inclusion $\forall w: A(w) \leq B(w)$	PSpace-complete				Undecidable		Open
Equivalence $\forall w: A(w) = B(w)$	PSpace-complete				Undecidable		Open

Kupferman and Lustig, “Lattice automata”, 2007.

Chatterjee, Doyen, and Henzinger, “Quantitative languages”, 2008.

Degorre, Doyen, Gentilini, Raskin, and Torunczyk, “Energy and mean-payoff games with imperfect information”, 2010.

Chatterjee, Doyen, Edelsbrunner, Henzinger, and Rannou, “Mean-payoff automaton expressions”, 2010.

Hunter, Pérez, and Raskin, “Mean-payoff games with partial observation”, 2014.

Deciding Safety and Liveness

	Inf	Sup	LimInf	LimSup	LimInfAvg	LimSupAvg	DSum
Safety-liveness decomposition	$O(1)$	PTime					$O(1)$
Deciding safety	$O(1)$	PSpace-complete			???		$O(1)$
Deciding liveness	PSpace-complete				???		???

Theorem.

A is safe $\Leftrightarrow A = A^*$

A is live $\Leftrightarrow A^* \geq T$

Deciding Safety and Liveness

	Inf	Sup	LimInf	LimSup	LimInfAvg	LimSupAvg	DSum
Safety-liveness decomposition	$O(1)$	PTime					$O(1)$
Deciding safety	$O(1)$	PSpace-complete			???		$O(1)$
Deciding liveness	PSpace-complete				???		???

Theorem.

A is safe $\Leftrightarrow A - A^* = 0$

A is live $\Leftrightarrow A^* = \top$

Given A and $k \in \mathbb{Q}$

$\forall w : A(w) < k$

$\forall w : A(w) = k$

$\forall w : A(w) \geq k$

PTime

???

Undecidable / Open
(limit-average) (disc. sum)

Constant-Function Check (CFC)

	Inf	Sup	LimInf	LimSup	LimInfAvg	LimSupAvg	DSum
Constant-function check	PSPACE-complete						

Theorem.

- NFA universality reduces to CFC
- Basic CFC reduces to basic universality
- Discounted-sum CFC reduces to NFA universality
- Limit-average CFC reduces to distance automata limitedness

Deciding Safety and Liveness

	Inf	Sup	LimInf	LimSup	LimInfAvg	LimSupAvg	DSum
Safety-liveness decomposition	$O(1)$	PTime					$O(1)$
Deciding safety	$O(1)$	PSpace-complete			ExpSpace; Pspace-hard		$O(1)$
Deciding liveness	PSpace-complete						
Constant-function check	PSpace-complete						

Recall.

$$A \text{ is safe} \iff A - A^* = 0$$



needs to be determinized

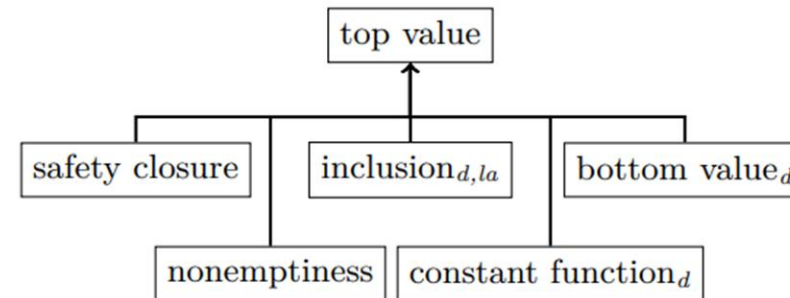
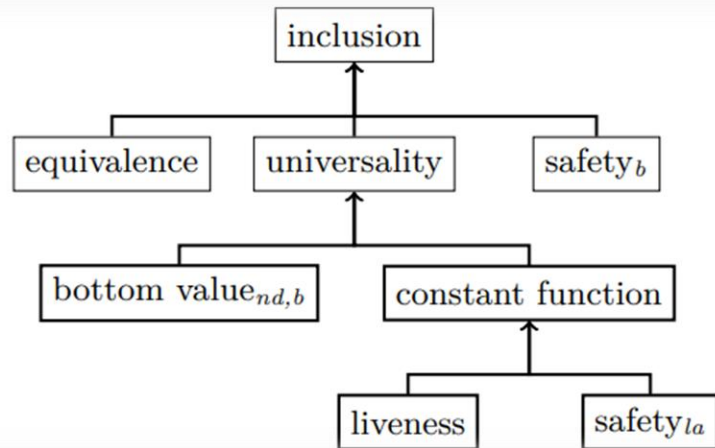
Part III

QuAK:

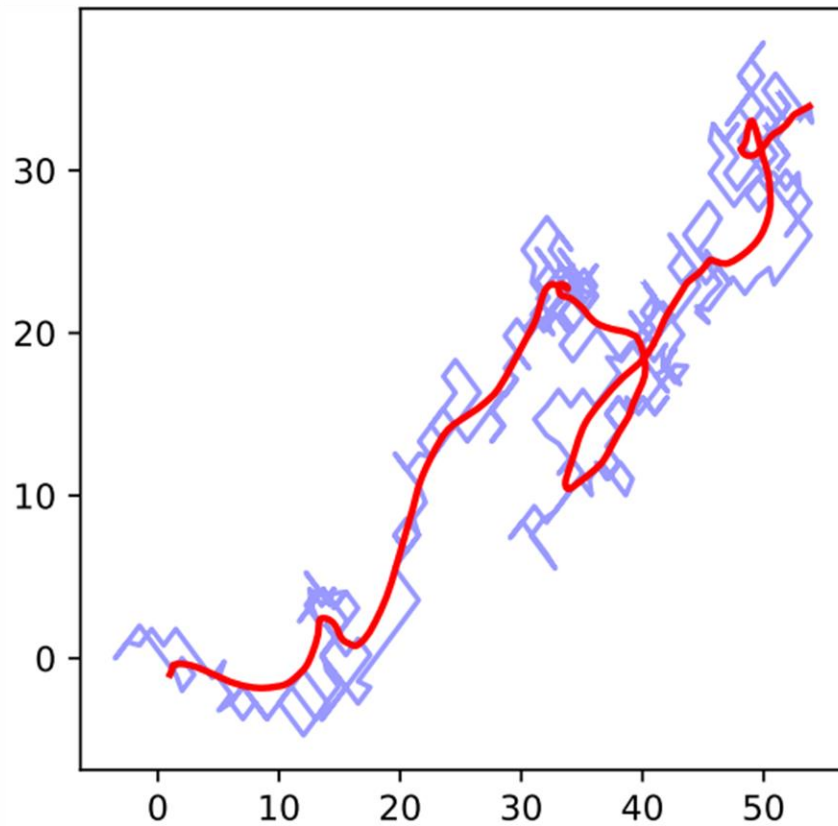
Quantitative Automata Kit

QuAK: Quantitative Automata Kit

- Implements both the standard and the novel algorithms
 - Emptiness, universality, inclusion...
 - Constant-check, safety closure, deciding safety/liveness...
 - Monitoring
- Available at github.com/ista-vamos/QuAK



Monitoring with QuAK



<i>Controller</i>	<u>Original</u>		<u>Smoothed</u>	
	<i>EC</i>	<i>Score</i>	<i>EC</i>	<i>Score</i>
sd=1	9.95	12.83	3.40	0.15
sd=5	12.42	15.97	2.42	0.82

Recap: Contributions

1. A theory of quantitative **monitorability** for reasoning about **resource-precision tradeoffs**
2. A **monitoring-oriented classification** of quantitative properties
3. A **software tool** for analyzing quantitative automata

Future Work

- An absolute notion of **resource use**
- **Synthesis** of approximate monitors
- Making quantitative safety and liveness **practical**
- Quantitative **hyperproperties**

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Future Work

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Contribution.

- A theory of quantitative **monitorability** for reasoning about **resource-precision tradeoffs**
- A **monitoring-oriented classification** of quantitative properties
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Future work.

- An absolute notion of **resource use**
- **Synthesis** of approximate monitors
- Making quantitative safety and liveness **practical**
- Quantitative **hyperproperties**

Thank you!

An Infinite Hierarchy of Limit Monitors

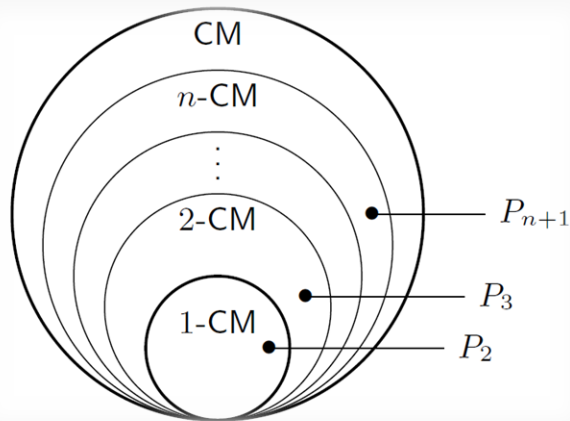
Theorem.

There is a quantitative property Φ such that

- Φ is not universally monitorable by any counter monitor
- each additional counter provides more precision

Idea:

Real-time counter hierarchy



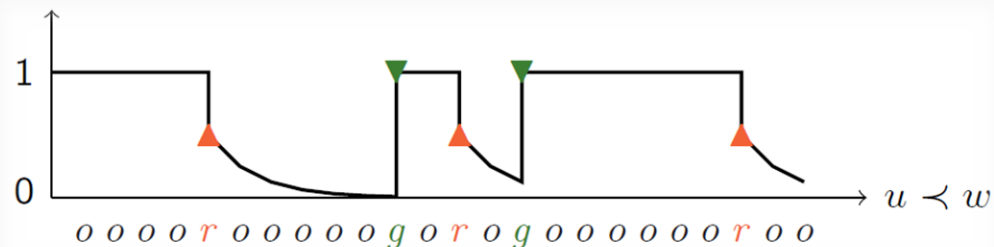
$P = \{ \#s_0\#s_1 \dots \mid \text{each } s_i \text{ encodes a natural number in binary, and} \\ \text{num}_i(u) \geq \text{num}_{i+1}(u) \text{ for every prefix } u \text{ and } i \in \mathbb{N} \}$

$\Phi(w) = \infty$, if $w \in P$
 $= |u|$, otherwise, where $u \prec w$ is the shortest bad prefix

Quantitative Prompt Monitoring with Abstract Monitors

Example (discounted response).

- $\Sigma = \{r, g, t, o\}$
- $\Phi(w) = \limsup_{u \prec w} \pi(u)$
- $\pi(u) = 1$, if every r in u is followed by some g
 $= \frac{1}{2^{n+1}}$, otherwise, where n is the number of observations since the earliest r without g



Quantitative Prompt Monitoring with Abstract Monitors

Example (abstract monitor for discounted response).

- $\Sigma = \{r, g, t, o\}$

- $M = (\sim, \gamma)$

a right-monotonic
equivalence relation on Σ^*

$\gamma: (\Sigma^* / \sim) \rightarrow \mathbb{D}$

Quantitative Prompt Monitoring with Abstract Monitors

Example (abstract monitor for discounted response).

- $\Sigma = \{r, g, t, o\}$

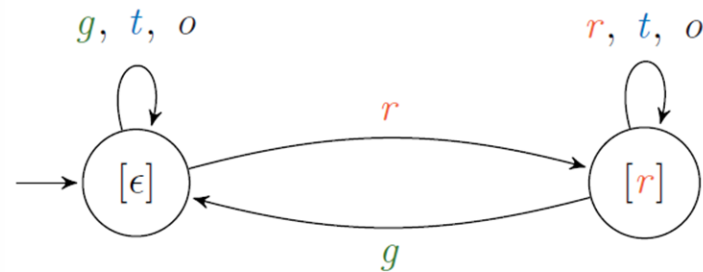
- $M = (\sim, \gamma)$

where $u_1 \sim u_2$ iff both have all requests granted or both have an open request, and

$$\gamma([u]) = 1, \text{ if all requests granted} \\ = 0, \text{ otherwise}$$

a right-monotonic
equivalence relation on Σ^*

$$\gamma: (\Sigma^*/\sim) \rightarrow \mathbb{D}$$



$$\gamma([ε]) = 1$$

$$\gamma([r]) = 0$$

Quantitative Prompt Monitoring with Abstract Monitors

Example (abstract monitor for discounted response).

- $\Sigma = \{r, g, t, o\}$

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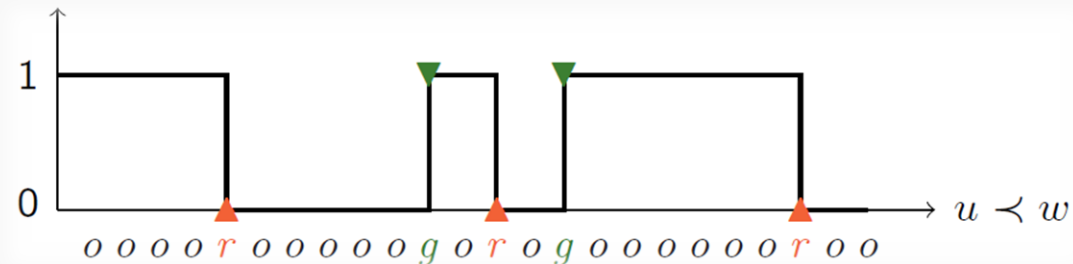
a right-monotonic
equivalence relation on Σ^*

$$\gamma: (\Sigma^* / \sim) \rightarrow \mathbb{D}$$

- M is a $\left(\frac{1}{2}, 0\right)$ -monitor for discounted response

worst-case error
on finite words

worst-case error
on infinite words



Quantitative Prompt Monitoring

with Abstract Monitors

Example (abstract monitor for discounted response).

- $\Sigma = \{r, g, t, o\}$
- $M' = (\sim', \gamma')$ where $u_1 \sim' u_2$ for every $u_1, u_2 \in \Sigma^*$
 $\gamma'([u]) = \frac{1}{2}$ for every $u \in \Sigma^*$
- M' is a $\left(\frac{1}{2}, \frac{1}{2}\right)$ -monitor for discounted response
- M' uses fewer resources than M (i.e., $\sim'$ grows more slowly than $\sim$)

An Infinite Hierarchy of Prompt Monitors

Theorem.

For all $i \in \mathbb{N}$ there is a monitor M_i such that

- M_i is an (i, i) -monitor for the maximal response-time property
- M_i uses fewer resources than M_j for all $j < i$

Idea:

Construct a monitor that counts response times precisely, but only stores an i -approximation of the maximum-so-far.

An Infinite Hierarchy of Prompt Monitors



An Infinite Hierarchy of Prompt Monitors

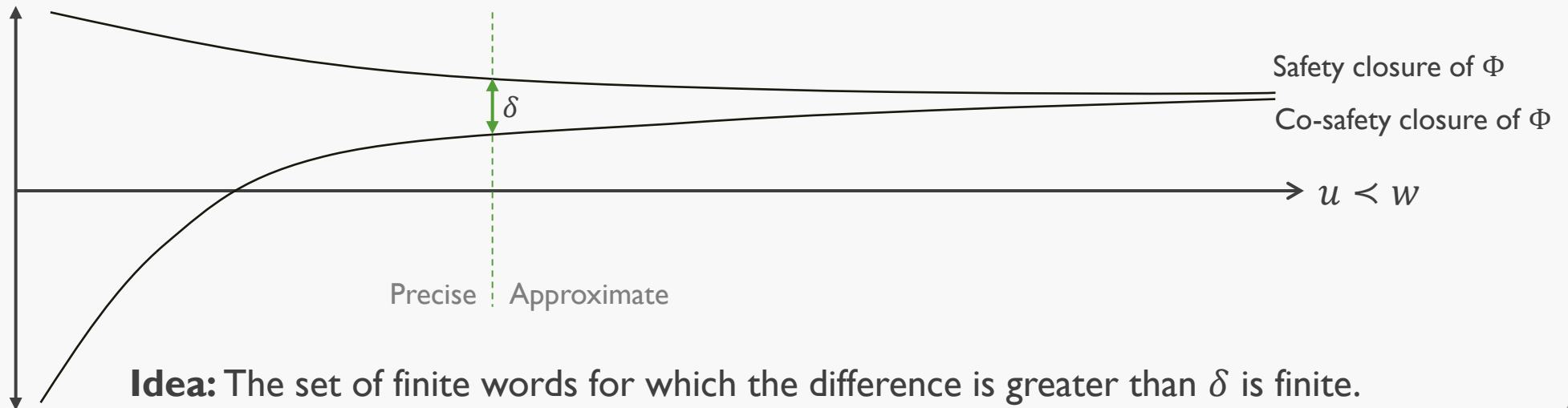


Finite-State Approximate Monitoring

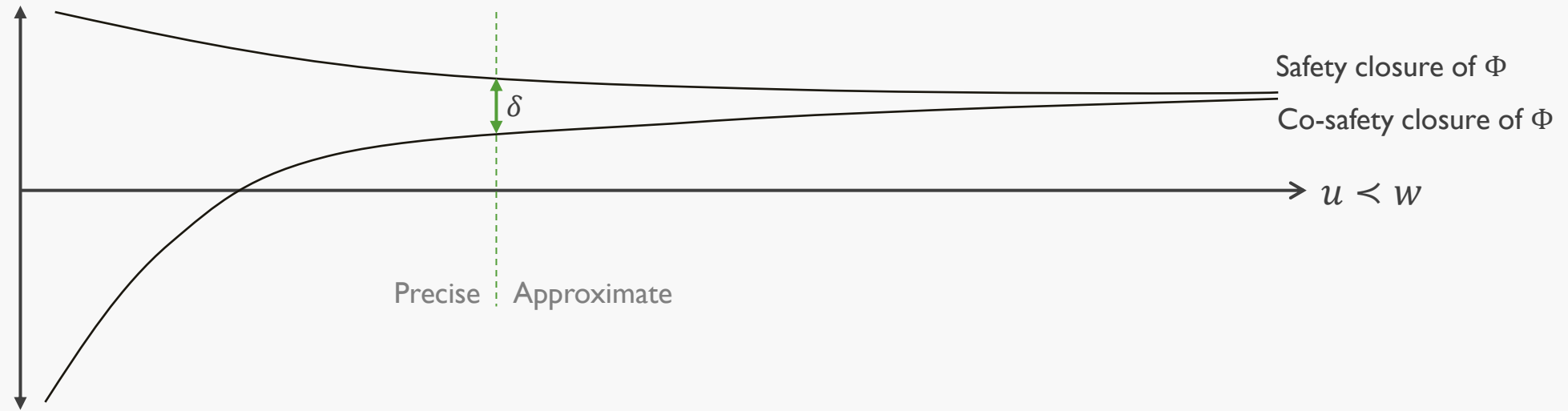
Theorem.

Φ is safe and co-safe
implies

for every $\delta > 0$ there is a *finite-state* (δ, δ) -monitor for Φ



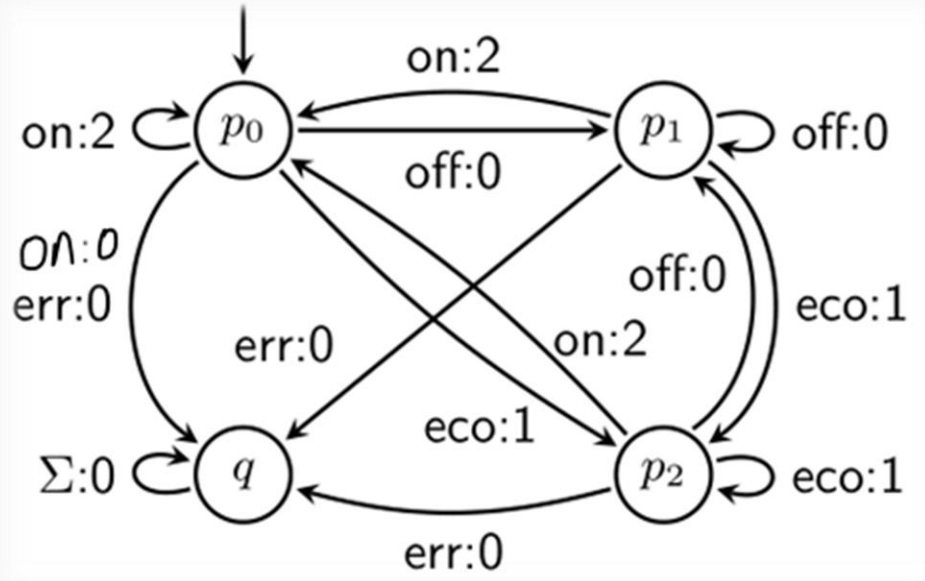
Finite-State Approximate Monitoring



Idea: The set of finite words for which the difference is greater than δ is finite.

- $S_\delta = \{u \in \Sigma^* : \sup_{x \in \Sigma^*} \pi(ux) - \inf_{y \in \Sigma^*} \pi(uy) \geq \delta\}$ is prefix closed
- By König's lemma there is $w \in \Sigma^\omega$ with all $u < w$ in S_δ
- Limit of suprema and infima over prefixes of w stays δ apart
- Contradiction to safety and co-safety

Quantitative Automata



$$w = \text{on} . \text{off} . (\text{eco} . \text{off})^\omega$$

$$s = 2 . 0 . (1 . 0)^\omega$$

- $\text{Sup}(s) = 2$
- $\text{LimSup}(s) = 1$
- $\text{LimInfAvg}(s) = \text{LimSupAvg}(s) = \frac{1}{2}$
- $\text{DSum}_2(s) = 2^0 \cdot 2 + 2^{-1} \cdot 0 + \dots = \frac{7}{3}$

$$A(w) = \sup\{v \mid v \text{ is the value of a run of } A \text{ on } w\}$$

- Acceptance condition $\rightarrow$ Value function
- $\text{Val} \in \{\text{Inf}, \text{Sup}, \text{LimInf}, \text{LimSup}, \text{LimInfAvg}, \text{LimSupAvg}, \text{DSum}\}$

Safety

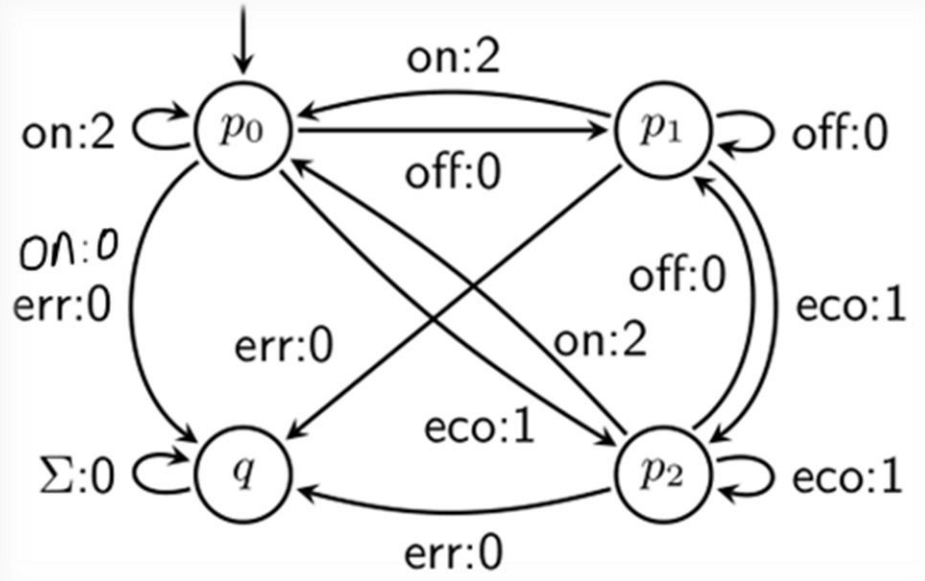
Reachability

Co-Büchi

Büchi

Games with quantitative objectives

Quantitative Automata



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Safety

Reachability

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Games with quantitative objectives

Constant-Function Check

	Inf	Sup	LimInf	LimSup	LimInfAvg	LimSupAvg	DSum
Constant-function check	PSpace-complete				PSpace-h.		PSpace-c.

Theorem.

- NFA universality reduces to constant-function check
- Discounted-sum constant-function check reduces to NFA universality

Constant-Function Check

	Inf	Sup	LimInf	LimSup	LimInfAvg	LimSupAvg	DSum
Constant-function check	PSpace-complete				PSpace-h.		PSpace-c.

Theorem.
Discounted-sum constant-function check reduces to
NFA universality.



$\Rightarrow$ NFA universal iff DSum constant T

Constant-Function Check

	Inf	Sup	LimInf	LimSup	LimInfAvg	LimSupAvg	DSum
Constant-function check	PSpace-complete				PSpace-c.		PSpace-c.

Theorem.

Limit-average constant-function check reduces to distance automata limitedness.

- Distance automata = Min-plus automata with weights in $\{0, 1\}$
- A distance automaton D is *limited* iff $\exists m \in \mathbb{N} : \forall u \in \Sigma^* : D(u) \leq m$

Constant-Function Check: Limit-Average

Limited $\Rightarrow$ Constant

- For every $w \in \Sigma^\omega$ there is an infinite run whose summation of weights are bounded.

Not limited $\Rightarrow$ Not constant

- There is $u \in \Sigma^*$ such that $D(u) = 1$ and the runs of D on u lead to a set of states S such that D^S is also not limited.
(D^S = the copy of D where the initial states are set to S)

Constant-Function Check: Limit-Average

Let A be a limit-average automaton with the top value T

- Construct a limit-average automaton B
 - only weights in $\{0, 1\}$
 - resolves nondeterminism with $\inf$
 - A is constant T iff B is constant 0
- Construct a distance automaton D from B by taking all states as accepting
 - Claim: B is constant 0 iff D is limited

Constant-Function Check: Limit-Average

Limited $\Rightarrow$ Constant

- Assume D is limited by some $b \in \mathbb{N}$.
- Let $w \in \Sigma^\omega$ be a word and consider the run tree T over its prefixes.
- There is an infinite path ρ where the sum of weights is bounded by b .
(Otherwise, there exists a finite prefix u of w with $D(u) > b$).
- Limit-average of such runs is 0.
- B maps all words to 0.

Constant-Function Check: Limit-Average

Not limited $\Rightarrow$ Not constant

(D^S = the copy of D where the initial states are set to S)

- Assume D is not limited.
- ($\star$): There is $u \in \Sigma^*$ such that $D(u) = 1$ and the runs of D on u lead to a set of states U such that D^U is also not limited.
- Repeatedly apply ($\star$) until reaching a set U_ℓ that was seen at j th iter.
- Take $w = u_1 \dots u_j (u_{j+1} \dots u_\ell)^\omega$ and let $m = \max_{1 \leq i \leq \ell} |u_i|$.
- For any run over w and for each i , the average after reading u_i is $\geq \frac{1}{m}$.
- B maps w to > 0 .